

Homework 5 (of 5) - solution

1. Basic concepts

1.1 What does the Central Limit Theorem tell us about the sampling distribution of the sample mean? How large should the sample size be to activate the Central Limit Theorem?

Answer: The Central Limit Theorem (CLT) states that, regardless of the shape of the population distribution, the sampling distribution of the sample mean will be approximately normally distributed, with a mean equal to the population mean and a standard deviation (standard error) equal to the population standard deviation divided by the square root of the sample size (that is, $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}}$).

For the CLT to "activate" (i.e., for the sampling distribution to become sufficiently normal):

- A sample size of 30 or more ($n \geq 30$) is the standard rule of thumb.
- If the population distribution is already approximately normal, the sampling distribution will be normal even with smaller sample sizes (e.g., $n < 30$).
- If the population distribution is already normal, the sampling distribution will be normal for any sample size (even for $n = 1$).

1.2 Describe the effect of increasing the sample size on the population of all possible sample proportions.

Answer: Increasing the sample size has two key effects on the population of all possible sample proportions:

1. Reduces the standard deviation: The standard deviation of the sampling distribution decreases as sample size increases. Its formula is $\sigma_{\hat{p}} = \sqrt{\frac{p(1-p)}{n}}$ (where p is the population proportion and n is sample size), so a larger n directly shrinks the spread—meaning sample proportions cluster more tightly around the true population proportion.
2. Improves normality (for the Central Limit Theorem): For non-extreme population proportions (e.g., p not too close to 0 or 1), a larger sample size makes the sampling distribution of the sample proportion more closely approximate a normal distribution. Even for extreme p , a bigger n can still satisfy the CLT's normality requirement (e.g., $np \geq 5$ and $n(1 - p) \geq 5$).
3. As n increases. It becomes easier to satisfy the $np \geq 5$ and $n(1 - p) \geq 5$ condition, forcing the distribution to become symmetric and bell-shaped.

1.3 For proportions, how large should the sample size be for being considered large?

Answer: $np \geq 5$ and $n(1 - p) \geq 5$, where p is the population proportion and n is sample size),

1.4 Explain why it is important to calculate a confidence interval

Answer: Calculating a confidence interval is important because it goes beyond a single point estimate (e.g., a sample mean or proportion) to quantify uncertainty—it tells you the range of values the true population parameter (e.g., true population mean or true population proportion) is likely to fall within, based on your sample data.

This uncertainty is critical because no sample is perfectly representative of its population (due to sampling variability). A confidence interval avoids the mistake of treating a point estimate as the "exact truth" and instead gives a realistic range, which is far more useful for decision-making (e.g., estimating customer satisfaction, drug effectiveness, or election results).

1.5 Under what conditions is the confidence interval $\left[\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$ for μ valid?

Answer: The confidence interval $\left[\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right]$ the population mean μ is valid only when three key conditions are met:

1. Random Sampling: The sample must be selected randomly from the population. This ensures the sample is representative and avoids bias that would invalidate the interval.
2. Known Population Standard Deviation (σ): The formula uses the population standard deviation (σ) (not the sample standard deviation, s). If σ is unknown, this z-interval is invalid, and a *t-interval* (using s) is required instead.
3. Normality of the Sampling Distribution: The sampling distribution of the sample mean ($\bar{x}$) must be approximately normal. This is satisfied if either:
 - The population itself is normally distributed (regardless of sample size), or
 - The sample size is large enough (typically $n \geq 30$) to activate the Central Limit Theorem (even for non-normal populations).

1.6 For a fixed sample size, what happens to a confidence interval for μ when we increase the level of confidence?

Answer: For a fixed sample size, increasing the level of confidence widens the confidence interval for μ .

This is because a higher confidence level (e.g., 99% vs. 95%) means we need to capture the true population mean μ with greater certainty. To do this, the critical z-value ($z_{\alpha/2}$) in the interval formula ($\left[\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right]$) increases (e.g., z for 95% confidence is ~ 1.96 , while for 99% it's ~ 2.576), which directly expands the margin of error ($z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$) and thus the entire interval.

1.7 For a fixed level of confidence, what happens to a confidence interval for μ when we increase the sample size?

Answer: For a fixed level of confidence, increasing the sample size narrows the confidence interval for μ .

This is because the interval's width depends on the margin of error, which includes the term $\frac{\sigma}{\sqrt{n}}$ (standard error). As sample size (n) increases, the standard error decreases—this shrinks the margin of error directly. Since the confidence level is fixed, the critical z-value ($z_{\alpha/2}$) doesn't change, so the smaller margin of error leads to a tighter interval around the sample mean ($\bar{x}$).

1.8 Explain how each of the following changes as *the number of degrees of freedom* describing a t curve increases:

a The standard deviation of the t curve. **b** The points t_α and $t_{\alpha/2}$.

Answer

a: As the number of degrees of freedom (df) describing a t curve increases, the standard deviation of the t curve decreases and approaches 1 (the standard deviation of the standard normal z -curve).

This is because t -curves are designed to account for uncertainty in estimating the population standard deviation (σ) with the sample standard deviation (s). With small df (e.g., $df = 5$), the t -curve is wider and more spread out (standard deviation > 1) to reflect greater uncertainty. As df increases (e.g., $df = 30$ or more), the sample standard deviation (s) becomes a more accurate estimate of σ , so the t -curve's spread shrinks and aligns closely with the z -curve (standard deviation = 1).

b: As the number of degrees of freedom (df) describing a t curve increases, both critical points t_α and $t_{\alpha/2}$ decrease in magnitude and approach the corresponding critical z -values (z_α and $z_{\alpha/2}$) from the standard normal distribution.

This happens because t -curves are wider and have fatter tails than the z -curve when df is small—so larger critical values (t_α and $t_{\alpha/2}$) are needed to capture the same tail area (α or $\alpha/2$). As df grows, the t -curve shrinks toward the z -curve (less spread, thinner tails), so smaller critical values suffice to capture the same tail area, eventually matching the z -values (e.g., for $\alpha/2 = 0.025$, $t_{\alpha/2}$ approaches 1.96 as df increases).

2.

7.20 When a pizza restaurant's delivery process is operating effectively, pizzas are delivered in an average of 45 minutes with a standard deviation of 6 minutes. To monitor its delivery process, the restaurant randomly selects five pizzas each night and records their delivery times.

a For the sake of argument, assume that the population of all delivery times on a given evening is normally distributed with a mean of $\mu = 45$ minutes and a standard deviation of $\sigma = 6$ minutes.

(That is, we assume that the delivery process is operating effectively.) Find the mean and the standard deviation of the population of all possible sample means, and calculate an interval containing 99.73 percent of all possible sample means.

b Suppose that the mean of the five sampled delivery times on a particular evening is $\bar{x} = 55$ minutes. Using the interval that you calculated in a, what would you conclude about whether the restaurant's delivery process is operating effectively? Why?

Answer

a

1. Mean of the sampling distribution of the sample mean ($\mu_{\bar{x}}$):

For any sampling distribution, the mean of all possible sample means equals the population mean. Thus, $\mu_{\bar{x}} = \mu = 45$ minutes.

2. Standard deviation of the sampling distribution (standard error, $\sigma_{\bar{x}}$):

Here the sample size is $n = 5$. $\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{5}} \approx 2.683$ minutes.

3. Interval containing 99.73% of all possible sample means:

For a normal distribution, 99.73% of data lies within 3 standard deviations of the mean (Empirical Rule).

Interval = $\mu_{\bar{x}} \pm 3\sigma_{\bar{x}} = 45 \pm 3 \times 2.683 \approx 45 \pm 8.049$, which is approximately (36.95 minutes, 53.05 minutes).

b Conclusion About Delivery Process

If the sampled mean $\bar{x} = 55$ minutes, we would conclude the delivery process is not operating effectively.

This is because 55 minutes falls outside the 99.73% interval (36.95, 53.05) calculated in part a. When the process is working effectively, 99.73% of all possible sample means should lie within this interval—observing a sample mean outside it is extremely unlikely (probability $\sim 0.27\%$), suggesting the true population mean (μ) may have increased (i.e., deliveries are slower).

3.

7.27 Suppose that we will randomly select a sample of $n = 100$ elements from a population and that we will compute the sample proportion $\hat{p}$ of these elements that fall into a category of interest. If the true population proportion p equals 0.9:

- a** Describe the shape of the sampling distribution of $\hat{p}$. Why can we validly describe the shape?
- b** Find the mean and the standard deviation of the sampling distribution of $\hat{p}$.

Answer

a The sampling distribution of $\hat{p}$ is approximately normal.

We can validly describe it this way because the success-failure condition (a key requirement for the Central Limit Theorem to apply to proportions) is satisfied by $np = 100(0.9) = 90 \geq 5$ and $n(1 - p) = 100(1 - 0.1) = 10 \geq 5$.

b Mean and Standard Deviation of the Sampling Distribution of $\hat{p}$

1. Mean of the sampling distribution ($\mu_{\hat{p}}$)

$$\mu_{\hat{p}} = p = 0.9.$$

2. Standard deviation of the sampling distribution (standard error, $\sigma_{\hat{p}}$):

$$\sigma_{\hat{p}} = \sqrt{\frac{p(1 - p)}{n}} = \sqrt{\frac{0.9(1 - 0.9)}{100}} = 0.03$$

4.

7.28 For the situation in Exercise 7.27, calculate the following probabilities. In each case sketch the sampling distribution and the probability.

a $P(\hat{p} \geq 0.96)$

b $P(0.855 \leq \hat{p} \leq 0.945)$

c $P(\hat{p} \leq 0.915)$

Answer

From problem 7.27, we have got $\mu_{\hat{p}} = 0.9$ and $\sigma_{\hat{p}} = 0.03$.

Note that $z = \frac{\hat{p} - \mu_{\hat{p}}}{\sigma_{\hat{p}}}$

a $P(\hat{p} \geq 0.96) = P\left(z \geq \frac{0.96 - 0.9}{0.03}\right) = P(z \geq 2).$

The standard normal table gives the area to the left of a z-score. From the z table, we can find the area to the **left** of $z = 2$ is 0.9772. Thus, area to the **right** of $z = 2$ is $P(z \geq 2) = 1 - P(z \leq 2) = 1 - 0.9772 = 0.0228$. Hence, $P(\hat{p} \geq 0.96) = 0.0228$.

b $P(0.855 \leq \hat{p} \leq 0.945) = P\left(\frac{0.855 - 0.9}{0.03} \leq z \leq \frac{0.945 - 0.9}{0.03}\right) = P(-1.5 \leq z \leq 1.5).$

The standard normal table gives the area to the left of a z-score. From the z table, we can find the area to the **left** of $z = -1.5$ is 0.0668, and the area to the left of $z = 1.5$ is 0.9332. Thus, area to between $z = -1.5$ and $z = 1.5$ is $0.9332 - 0.0668 = 0.8664$. Hence, $P(0.855 \leq \hat{p} \leq 0.945) = 0.8664$.

c $P(\hat{p} \leq 0.915) = P\left(z \leq \frac{0.915 - 0.9}{0.03}\right) = P(z \leq 0.5).$

The standard normal table gives the area to the left of a z-score. From the z table, we can find the area to the **left** of $z = 0.5$ is 0.6915. Hence, $P(\hat{p} \leq 0.915) = 0.6915$.

5.

8.8 THE BANK CUSTOMER WAITING TIME CASE

Recall that a bank manager has developed a new system to reduce the time customers spend waiting to be served by tellers during peak business hours. The mean waiting time during peak business hours under the current system is roughly 9 to 10 minutes. The bank manager hopes that the new system will have a mean waiting time that is less than six minutes. The mean of the sample of 100 bank customer waiting times in Table 1.8 is $\bar{x} = 5.46$. If we let μ denote the mean of all possible bank customer waiting times using the new system and assume that σ equals 2.47:

- a** Calculate 95 percent and 99 percent confidence intervals for μ .
- b** Using the 95 percent confidence interval, can the bank manager be 95 percent confident that μ is less than six minutes? Explain.
- c** Using the 99 percent confidence interval, can the bank manager be 99 percent confident that μ is less than six minutes? Explain.
- d** Based on your answers to parts b and c, how convinced are you that the new mean waiting time is less than six minutes?

Answer

a: Since n is larger than 30 and σ is known, we can use the z-interval formula $\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$

For 95% Confidence Interval

Critical z-value ($z_{\alpha/2}$) for 95% confidence is 1.96 (this can be found from the z table). So, the interval is $\left[\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] = \left[5.46 - 1.96 \frac{2.47}{\sqrt{100}}, 5.46 + 1.96 \frac{2.47}{\sqrt{100}} \right] = [4.976, 5.944]$.

For 99% Confidence Interval

Critical z-value ($z_{\alpha/2}$) for 99% confidence is 2.576 (this can be found from the z table). So, the interval is $\left[\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] = \left[5.46 - 2.576 \frac{2.47}{\sqrt{100}}, 5.46 + 2.576 \frac{2.47}{\sqrt{100}} \right] = [4.823, 6.097]$.

b: For 95% Confidence Interval: Yes, the bank manager can be 95% confident that μ is less than six minutes.

The 95% interval is (4.976, 5.944), and the entire interval is below 6 minutes. This means we are 95% certain the true mean waiting time falls within a range that is entirely less than 6, so μ must be less than 6.

c: For 99% Confidence Interval: No, the bank manager cannot be 99% confident that μ is less than six minutes.

The 99% interval is (4.823, 6.097), and the upper bound of the interval (6.097) is above 6 minutes. This means there is a 1% chance the true mean could be as high as ~ 6.1 minutes, so we cannot rule out $\mu \geq 6$ with 99% confidence.

d: We are moderately (95%) to highly convinced (99%) the new mean waiting time is less than 6 minutes.

The 95% interval (a common confidence level for practical decisions) entirely excludes 6, providing strong evidence for $\mu < 6$. While the 99% interval (a more conservative level) includes values slightly above 6, the lower bound (4.823) is well below 6, and the upper bound (6.097) is only marginally above—suggesting the risk of $\mu \geq 6$ is very low, even if not eliminated at 99% confidence.

6.

8.16 Suppose that for a sample of $n = 11$ measurements, we find that $\bar{x} = 72$ and $s = 5$. Assuming normality, compute confidence intervals for the population mean μ with the following levels of confidence:

a 95% b 99% c 80% d 90% e 98% f 99.8%

Answer

To compute the confidence intervals for μ , we use the t -interval formula (since the population standard deviation σ is unknown, we use the sample standard deviation s , and the population is normal). The degree of freedom is $df = n - 1 = 11 - 1 = 10$.

For each confidence level, we first find the critical t -value ($t_{\alpha/2}$ with $df = 10$) from the t -table. We then calculate interval using the formula $\left[\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right]$.

a. 95% Confidence Interval

$z_{0.025}$ with $df = 10$ is 2.228. Interval is $\left[\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right] = \left[72 - 2.228 \frac{5}{\sqrt{11}}, 72 + 2.228 \frac{5}{\sqrt{11}} \right] = [68.64, 75.36]$

b. 99% Confidence Interval

$z_{0.005}$ with $df = 10$ is 3.169. Interval is $\left[\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right] = \left[72 - 3.169 \frac{5}{\sqrt{11}}, 72 + 3.169 \frac{5}{\sqrt{11}} \right] = [67.22, 76.78]$

c. 80% Confidence Interval

$z_{0.1}$ with $df = 10$ is 1.3729. Interval is $\left[\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right] = \left[72 - 1.372 \frac{5}{\sqrt{11}}, 72 + 1.372 \frac{5}{\sqrt{11}} \right] = [69.63, 74.07]$

d. 90% Confidence Interval

$z_{0.05}$ with $df = 10$ is 1.812. Interval is $\left[\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right] = \left[72 - 1.812 \frac{5}{\sqrt{11}}, 72 + 1.812 \frac{5}{\sqrt{11}} \right] = [69.27, 74.73]$

e. 98% Confidence Interval

$z_{0.01}$ with $df = 10$ is 2.764. Interval is $\left[\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right] = \left[72 - 2.764 \frac{5}{\sqrt{11}}, 72 + 2.764 \frac{5}{\sqrt{11}} \right] = [68.83, 76.17]$

f. 99.8% Confidence Interval

$z_{0.001}$ with $df = 10$ is 4.140. Interval is $\left[\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}} \right] = \left[72 - 4.140 \frac{5}{\sqrt{11}}, 72 + 4.140 \frac{5}{\sqrt{11}} \right] = [65.76, 78.24]$

7.

8.28 Consider a population having a standard deviation equal to 10. We wish to estimate the mean of this population.

a How large a random sample is needed to construct a 95 percent confidence interval for the mean of this population with a margin of error equal to 1?

b Suppose that we now take a random sample of the size we have determined in part a. If we obtain a sample mean equal to 295, calculate the 95 percent confidence interval for the population mean. What is the interval's margin of error?

Answer

To find the required sample size, we use the margin of error formula for z -intervals (since σ is known): $E = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$. So, $n = \left(\frac{z_{\alpha/2} \times \sigma}{E} \right)^2$. We also know that $\sigma = 10$.

a: For confidence level = 95%, from the z table, we can find the z -value ($z_{\alpha/2} = z_{0.025}$) = 1.96.

So, $n = \left(\frac{z_{\alpha/2} \times \sigma}{E} \right)^2 = \left(\frac{1.96 \times 10}{0.1} \right)^2 = 19.6^2 = 384.16$. Round up to $n = 385$.

b: For confidence interval for sample mean $\bar{x} = 295$ and sample size $n = 385$.

$$\left[\bar{x} \pm z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \right] = \left[295 - 1.96 \frac{10}{\sqrt{385}}, 295 + 1.96 \frac{10}{\sqrt{385}} \right] = [294, 296]$$

8.

8.38 In a news story distributed by the *Washington Post*, Lew Sichelman reports that a substantial fraction of mortgage loans that go into default within the first year of the mortgage were approved on the basis of falsified applications. For instance, loan applicants often exaggerate their income or fail to declare debts. Suppose that a random sample of 1,000 mortgage loans that were defaulted within the first year reveals that 410 of these loans were approved on the basis of falsified applications.

a Find a point estimate of and a 95 percent confidence interval for p , the proportion of all first-year defaults that are approved on the basis of falsified applications.

b Based on your interval, what is a reasonable estimate of the minimum percentage of all first-year defaults that are approved on the basis of falsified applications?

Answer

a. The point estimate of the population proportion p is the sample proportion $\hat{p}$, which is calculated as:

$$\hat{p} = \frac{\text{number of applications}}{\text{total sample size}} = \frac{410}{1000} = 0.41$$

Since $n\hat{p} = 1000(0.41) = 410 > 5$ and $n(1 - \hat{p}) = 1000(1 - 0.41) = 590 > 5$, we can use the z -interval

for proportions. The interval is $\left[\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right]$

For 95% confidence, we can find from the z table that $(z_{\alpha/2} = z_{0.025}) = 1.96$. So, the Confidence Interval is:

$$\left[\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} \right] = \left[0.41 \pm 1.96 \sqrt{\frac{0.41(1-0.41)}{1000}} \right] = [0.3794, 0.4406].$$

b. The 95% confidence interval for p is (0.3794, 0.4406), which translates to 37.94% to 44.06%. A reasonable estimate of the minimum percentage of all first-year defaults approved via falsified applications is the lower bound of the 95% confidence interval, approximately 38% (rounded to a practical whole number).

9.

14.7 An accountant wishes to predict direct labor cost (y) on the basis of the batch size (x) of a product produced in a job shop. Data for 12 production runs are given in the table.

- Which ones are the dependent variable and the independent variable?
- Draw a scatter diagram with the independent variable on the horizontal axis and the dependent variable on the y axis.
- Determine the linear regression equation $\hat{y} = b_0 + b_1x$.
- Calculate the coefficient of correlation (r) between the two variables

Direct Labor Cost
Data 

Direct Labor Cost, y (\$100s)	Batch Size, x
71	5
663	62
381	35
138	12
861	83
145	14
493	46
548	52
251	23
1024	100
435	41
772	75